

Notes: Barroso, Detzel, and Maio (JFE, 2025)

The Volatility Puzzle of the Beta Anomaly

Contents

1	Big picture	2
2	Data and variable construction	2
2.1	Data sources	2
2.2	Estimating stock betas	3
2.3	Constructing the BAB factor	3
2.4	Realized volatility and volatility-managed BAB	3
2.5	Conditioning variable	4
3	Models and empirical specifications	4
3.1	Volatility management test	4
3.2	Leverage-constraints model implication	4
3.3	Factor-model spanning regressions	5
3.4	Conditional factor models	5
3.5	Limits to arbitrage tests	5
3.6	Lottery demand tests	5
3.7	Sentiment and disagreement tests	6
3.8	Demand system	6
3.9	Counterfactual repricing experiment	6
4	Identification and empirical design	7
4.1	What identifies the volatility puzzle	7
4.2	Why the volatility state is informative	7
4.3	Theory-by-theory falsification logic	7
4.4	Identification in the demand system	7
4.5	Limitations	8
5	Tables and figures	9
5.1	Figure 1 and Table 1: BAB volatility and volatility-managed performance	9
5.2	Tables 2 and 3: leverage constraints and conditional BAB performance	9
5.3	Table 4: conditional factor models	10
5.4	Tables 5 and 6: IVOL, mispricing, and lottery demand	10
5.5	Table 7: sentiment and disagreement	11
5.6	Table 8: BAB volatility versus market volatility	11
5.7	Tables 9 and 10: institutional demand and counterfactual repricing	13
6	Short summary	13
7	Conclusion	14
7.1	Core conclusion	14
7.2	Economic intuition	14
7.3	Takeaway	14

1 Big picture

- **Question.** Can leading explanations of the beta anomaly explain not only its unconditional average return but also its conditional performance across high- and low-volatility states?
- **Object of study.** The paper studies betting-against-beta (BAB) factors that buy low-beta stocks and short high-beta stocks, while hedging market exposure to target beta neutrality.
- **Central empirical fact.** BAB abnormal returns and Sharpe ratios are much higher after months with below-median realized BAB volatility than after months with above-median volatility.
- **Puzzle.** Existing theories can explain average BAB returns, but they fail to explain why BAB is most profitable and most anomalous when its realized volatility is low.
- **Leverage-constraints theory test.** The Frazzini–Pedersen leverage-constraints model counterfactually predicts that market and BAB Sharpe ratios should rise with volatility. The data show the opposite for BAB.
- **Multifactor-model test.** The Fama–French six-factor model prices BAB unconditionally, but its factor loadings shrink or flip in low-volatility states, leaving a large positive conditional alpha.
- **Limits-to-arbitrage test.** Dropping high-IVOL stocks, dropping overpriced stocks, or orthogonalizing beta to IVOL/mispricing removes unconditional alpha, but the low-minus-high volatility alpha remains large and significant.
- **Lottery, sentiment, and disagreement tests.** Orthogonalizing beta to lottery-demand proxies, conditioning on sentiment, or conditioning on analyst disagreement also does not explain the volatility pattern.
- **Demand-side mechanism.** Institutional investors prefer high-beta stocks on average, but this preference collapses when volatility is high. They shift demand toward low-beta stocks as volatility rises.
- **Counterfactual price impact.** Dropping institutions whose portfolio betas fall most strongly when volatility increases causes low-beta stocks to appreciate relative to high-beta stocks, with a state-dependent price impact large enough to explain the conditional BAB alpha.
- **Economic takeaway.** The beta anomaly is not fully explained by static theories of leverage constraints, limits to arbitrage, lottery demand, sentiment, disagreement, or standard factor exposure. Its conditional performance points to time-varying institutional demand for beta.

2 Data and variable construction

2.1 Data sources

- The sample period is July 1965 to December 2016, constrained by the availability of the variables required for the full set of tests.
- Stock returns come from CRSP, accounting data from Compustat, institutional holdings from Thomson Reuters 13F data, and analyst forecasts from I/B/E/S.
- Factor returns come from Kenneth French’s data library, including the Fama–French six-factor model: market, size, value, profitability, investment, and momentum.
- Mispricing scores come from Stambaugh, Yu, and Yuan’s anomaly-based mispricing measure.
- Sentiment comes from Baker and Wurgler’s sentiment index orthogonalized to macroeconomic conditions.
- Consumption growth and intermediary leverage variables are used for robustness factor models.

Interpretation: The paper deliberately combines factor data, stock-level anomaly measures, analyst disagreement, institutional holdings, and demand-system estimates. This allows the authors to test multiple existing explanations under the same conditioning variable: lagged BAB volatility.

2.2 Estimating stock betas

For each stock-month (n, t) , the paper estimates CAPM regressions using the previous 60 monthly returns, with a minimum of 36 observations, and a Dimson correction for nonsynchronous trading:

$$r_{n,\tau} - r_{f,\tau} = a_{n,t} + \beta_{n,t,0}\text{MKT}_\tau + \beta_{n,t,1}\text{MKT}_{\tau-1} + \epsilon_\tau, \quad \tau = t - 59, \dots, t. \quad (1)$$

The time-series beta estimate is

$$\hat{\beta}_{n,t}^{ts} = \hat{\beta}_{n,t,0} + \hat{\beta}_{n,t,1}. \quad (2)$$

The paper then shrinks beta toward one using a Vasicek-style procedure:

$$\hat{\beta}_{n,t} = w_n \hat{\beta}_{n,t}^{ts} + (1 - w_n) \times 1, \quad (3)$$

where w_n is increasing in the precision of the time-series beta estimate and decreasing in the cross-sectional dispersion of true betas.

Interpretation: The beta estimates are conservative because the shrinkage reduces estimation error and pulls extreme beta estimates toward the market beta of one.

2.3 Constructing the BAB factor

At the beginning of each month, stocks are sorted into value-weighted beta decile portfolios. Let $r_{L,t}$ be the return on the low-beta decile and $r_{H,t}$ be the return on the high-beta decile. The paper defines BAB as

$$\text{BAB}_t = r_{L,t} - r_{H,t} - (\beta_{L,t-1} - \beta_{H,t-1})\text{MKT}_t. \quad (4)$$

- The first two terms buy low-beta stocks and short high-beta stocks.
- The third term hedges the market exposure of the long-short portfolio.
- The hedge is crucial because unhedged low-minus-high beta portfolios can have positive alpha but insignificant mean returns.

Interpretation: The market hedge converts the beta anomaly into a factor with positive mean returns and near-zero market beta, making Sharpe ratios meaningful.

2.4 Realized volatility and volatility-managed BAB

Realized volatility of BAB in month t is computed from daily BAB returns:

$$\hat{\sigma}_t = \sqrt{\sum_{d=1}^{21} \text{BAB}_{d,t}^2}. \quad (5)$$

The volatility-managed BAB factor is

$$\text{BAB}_{t+1}^\sigma = \left(\frac{c}{\hat{\sigma}_t} \right) \text{BAB}_{t+1}, \quad (6)$$

where c is chosen so that the managed and unmanaged factors have the same unconditional volatility.

Interpretation: If the conditional Sharpe ratio of BAB is higher after low-volatility months, scaling up after low volatility and scaling down after high volatility should improve the Sharpe ratio.

2.5 Conditioning variable

The main conditioning variable is whether the previous month realized BAB volatility is above or below its sample median. The paper reports:

- unconditional estimates;
- estimates in low-volatility months;
- estimates in high-volatility months;
- the difference between low- and high-volatility states.

Interpretation: The logic is simple: a good theory of the beta anomaly must explain not only why BAB earns average returns, but also why it earns much stronger abnormal returns after low-volatility months.

3 Models and empirical specifications

3.1 Volatility management test

The volatility-management test asks whether

$$\text{SR}(\text{BAB}^\sigma) > \text{SR}(\text{BAB}). \quad (7)$$

The managed strategy has the same unconditional volatility as unmanaged BAB, so the comparison isolates whether the timing rule raises the average return per unit of risk.

Interpretation: A higher Sharpe ratio for BAB^σ indicates that BAB's conditional expected return does not rise enough in high-volatility states to compensate for its higher risk. This creates the basic volatility puzzle.

3.2 Leverage-constraints model implication

The Frazzini–Pedersen logic implies that leverage-constrained investors bid up high-beta stocks because they cannot lever low-beta assets. In equilibrium, expected excess returns satisfy a relation of the form

$$E(r_{n,t+1}) - r_f = \psi_t(1 - \beta_{n,t}) + \beta_{n,t}[E(r_{M,t+1}) - r_f], \quad (8)$$

where ψ_t is the aggregate shadow cost of leverage constraints.

- If $\psi_t > 0$, high-beta stocks have negative alphas and low-beta stocks have positive alphas.
- Because investors in the model are risk averse, expected returns and Sharpe ratios should rise with volatility.
- The paper calibrates the model under different volatility scalars and risk-aversion assumptions.

Interpretation: The model can generate the unconditional beta anomaly, but it predicts the wrong conditional relationship between volatility and Sharpe ratios.

3.3 Factor-model spanning regressions

The paper estimates regressions of BAB returns on factor models:

$$\text{BAB}_t = \alpha + \beta' \mathbf{f}_t + \epsilon_t. \quad (9)$$

It reports results for the CAPM, Fama–French three-factor model, and Fama–French six-factor model. It also estimates the same regressions separately in low- and high-volatility regimes.

Interpretation: If the FF6 explains the beta anomaly, BAB alpha should be small both unconditionally and conditionally. Instead, the unconditional FF6 alpha is near zero, but the low-volatility alpha is large and positive.

3.4 Conditional factor models

The paper estimates conditional factor models of the form

$$\text{BAB}_t = \alpha_0 + \alpha_1' \mathbf{z}_{t-1} + \sum_{k=1}^K (\gamma_{k,0} + \gamma_{k,1}' \mathbf{z}_{t-1}) f_{k,t} + \epsilon_t, \quad (10)$$

where $\mathbf{z}_{t-1}$ includes lagged volatility and other state variables.

Interpretation: This specification allows both the alpha and the factor loadings of BAB to vary with conditioning information. Even under these more flexible models, lagged volatility remains a strong negative predictor of abnormal BAB returns.

3.5 Limits to arbitrage tests

The paper builds five modified BAB factors:

1. dropping high-IVOL stocks;
2. dropping overpriced stocks;
3. dropping overpriced high-IVOL stocks;
4. sorting on beta orthogonalized to IVOL;
5. sorting on beta orthogonalized to mispricing.

For example, orthogonalizing beta to IVOL uses a cross-sectional regression each month:

$$\hat{\beta}_{n,t} = a_t + b_t \text{IVOL}_{n,t} + \nu_{n,t}, \quad (11)$$

then sorts stocks using the residual $\nu_{n,t}$.

Interpretation: These tests directly target the Liu–Stambaugh–Yuan explanation that beta is correlated with IVOL and mispricing. The conditional alpha remains, so IVOL/mispricing do not explain the volatility puzzle.

3.6 Lottery demand tests

Lottery demand is proxied by extreme positive daily returns:

- SMAX: average of the highest five daily returns in the month;
- MAX: maximum daily return in the month.

The paper forms BAB factors using beta residuals after orthogonalizing beta to these lottery-demand proxies.

Interpretation: If high-beta stocks underperform because they are lottery-like, controlling for lottery demand should remove conditional BAB alpha. It does not.

3.7 Sentiment and disagreement tests

The paper conditions BAB performance jointly on volatility and:

- Baker–Wurgler sentiment;
- analyst forecast disagreement.

Interpretation: Sentiment and disagreement theories imply the anomaly should be stronger when optimism or disagreement is high. The paper finds that BAB alpha remains strongly related to volatility within sentiment and disagreement states.

3.8 Demand system

The institutional demand analysis follows Kojen and Yogo. Investor i 's portfolio weight in stock n at time t is

$$w_{i,t}(n) = \frac{\delta_{i,t}(n)}{1 + \sum_{m=1}^N \delta_{i,t}(m)}. \quad (12)$$

The normalized weight relative to the outside asset is

$$\delta_{i,t}(n) = \exp \left(\theta_{i,t}^{me} me_t(n) + \theta_{i,t}^{\beta} \beta_t(n) + \boldsymbol{\theta}_t^{x'} \mathbf{x}_t(n) + \theta_{0,i,t} \right) \epsilon_{i,t}(n). \quad (13)$$

- $me_t(n)$ is log market equity.
- $\beta_t(n)$ is the stock beta.
- $\mathbf{x}_t(n)$ contains controls such as book equity, profitability, investment, and dividends to book equity.
- $\theta_{i,t}^{\beta}$ measures the institution's demand tilt toward high-beta stocks.

Interpretation: Estimating investor-specific beta demand allows the paper to ask whether institutions reduce their demand for high-beta stocks as volatility rises.

3.9 Counterfactual repricing experiment

Each quarter, institutions are ranked by the sensitivity of their portfolio beta to BAB volatility:

$$\beta_{i,\tau} = a_{i,t} + b_{i,t} \hat{\sigma}_{\tau-1} + \nu_{i,\tau}, \quad \tau = t - 40, \dots, t - 1. \quad (14)$$

The experiment drops the institutions with the most negative $b_{i,t}$ coefficients until they account for 10% of aggregate assets under management, then reallocates their assets to the remaining institutions. The resulting price impact for portfolio p is

$$PI_t(p) = \frac{ME_t^{CF}(p) - ME_t(p)}{ME_t(p)}. \quad (15)$$

Interpretation: This experiment identifies institutions most prone to reducing beta exposure when volatility rises and asks whether their demand dynamics can account for the conditional BAB alpha.

4 Identification and empirical design

4.1 What identifies the volatility puzzle

- The key variation is time-series variation in realized BAB volatility.
- The paper compares BAB returns, factor alphas, and Sharpe ratios after low- and high-volatility months.
- Existing theories are evaluated by whether they can explain this conditional pattern.

Interpretation: The paper’s empirical discipline is conditional performance. A theory that explains only average BAB returns but fails across volatility states is rejected or judged incomplete.

4.2 Why the volatility state is informative

- Volatility is observable and persistent enough to condition on without look-ahead bias.
- Volatility is a natural state variable for risk, arbitrage limits, disagreement, and institutional risk-management incentives.
- Volatility-managed factor evidence already suggests that many factor premia have state-dependent Sharpe ratios.

Interpretation: If risk compensation or constraints explain the beta anomaly, high volatility should be a natural place for those explanations to show up. Instead, BAB is strongest when volatility is low.

4.3 Theory-by-theory falsification logic

- **Leverage constraints:** should imply a positive risk-return tradeoff and higher Sharpe ratios with volatility; data show the opposite.
- **Multifactor exposures:** should absorb BAB alpha in both states; FF6 only works unconditionally.
- **Limits to arbitrage:** should remove alpha after controlling for IVOL and mispricing; the conditional alpha remains.
- **Lottery demand:** should disappear after orthogonalizing beta to SMAX or MAX; it does not.
- **Sentiment/disagreement:** should dominate volatility if optimism or disagreement drives the anomaly; volatility remains central.

Interpretation: The paper’s contribution is not merely another explanation of the beta anomaly. It shows that several plausible explanations fail a common conditional test.

4.4 Identification in the demand system

- Institutional holdings identify cross-sectional and time-series patterns in demand for stock characteristics.
- The Kojen–Yogo instrument uses other investors’ holdings to address the endogeneity of market equity.
- Demand coefficients are estimated by institution type and by volatility state.
- Counterfactual repricing uses market clearing to translate demand changes into price changes.

Interpretation: The institutional analysis provides a mechanism for the conditional pattern: institutions reduce high-beta demand when volatility rises, causing relative prices of low- and high-beta stocks to move in a way that accounts for BAB’s state dependence.

4.5 Limitations

- Conditional tests are strong diagnostic tools but are not randomized experiments.
- BAB volatility can reflect many economic forces, so interpretation requires theory-specific auxiliary tests.
- Demand-system estimates rely on holdings data, asset-pricing controls, and market-clearing assumptions.
- Household demand is measured residually and should be interpreted with caution.
- The counterfactual experiment establishes plausibility and magnitude, not a literal policy experiment.

Interpretation: The paper’s strongest claim is comparative: leading explanations fail the volatility-conditioning test, while institutional demand dynamics produce price impacts of the right sign and magnitude.

5 Tables and figures

Visuals are directly extracted from the paper and grouped compactly.

5.1 Figure 1 and Table 1: BAB volatility and volatility-managed performance

Read:

- Figure 1 shows substantial time variation in monthly realized volatility of the BAB factor.
- Table 1 compares unmanaged BAB to volatility-managed BAB.
- BAB earns a mean return of 0.45% per month with a Sharpe ratio of 0.34.
- Volatility-managed BAB has the same standard deviation but a higher mean return of 0.67% and a higher Sharpe ratio of 0.50.
- Volatility management also reduces skewness and kurtosis relative to unmanaged BAB.

Economic message: BAB’s conditional Sharpe ratio is higher after low-volatility months. This motivates the entire paper: any explanation of the beta anomaly must explain why the anomaly is strongest when BAB volatility is low.

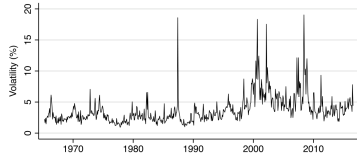


Figure 1: Realized volatility of BAB

Table 1

Performance of BAB and its volatility-managed counterpart.

	Mean (%)	SD (%)	Skew	Kurt	SR
BAB	0.45	4.64	−0.91	8.06	0.34
BAB*	0.67	4.64	0.01	3.68	0.50
					(2.23)

Table 1: BAB and volatility-managed BAB

5.2 Tables 2 and 3: leverage constraints and conditional BAB performance

Read:

- Table 2 calibrates the Frazzini–Pedersen leverage-constraints model under alternative volatility scalars.
- The model predicts that market and BAB Sharpe ratios increase as volatility increases.
- Table 3 shows the empirical opposite: BAB’s FF6 alpha is 0.56% in low-volatility months and −0.41% in high-volatility months.
- The low-minus-high alpha spread is 0.97% per month.
- BAB’s Sharpe ratio is 0.72 in low-volatility months and 0.13 in high-volatility months.

Economic message: The leverage-constraints theory explains average BAB returns but predicts the wrong conditional risk-return tradeoff.

Table 2

Calibrations of the Frazzini and Pedersen (2014) leverage constraints model.

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Exogenous variables								
Volatility scalar	1	1	1	0.25	0.5	1.5	2	2
Risk aversion 1	1	1	1	1	1	1	1	1
Risk aversion 2	1	10	10	10	10	10	10	10
Wealth share 1	0.5	0.5	0.8	0.8	0.8	0.8	0.8	0.8
m1	1	1	1.2	1.2	1.2	1.2	1.2	1.5
m2	1	1	0	0	0	0	0	0
Endogenous variables								
Vol L	13%	14%	14%	7%	9%	18%	21%	22%
Vol H	30%	33%	33%	15%	22%	42%	51%	55%
Vol MKT	21%	23%	23%	11%	15%	29%	35%	37%
Vol BAB	7%	7%	7%	3%	5%	9%	11%	12%
Excess return L	3%	9%	10%	3%	6%	14%	17%	26%
Excess return H	6%	16%	15%	4%	8%	22%	28%	37%
Excess return MKT	4%	12%	13%	3%	7%	18%	22%	31%
Excess return BAB	0%	4%	6%	2%	3%	8%	9%	16%
SR MKT	0.20	0.55	0.55	0.33	0.44	0.61	0.64	0.84
SR BAB	0.00	0.47	0.79	0.51	0.66	0.81	0.78	1.32
β_L	0.6	0.6	0.6	0.6	0.6	0.6	0.6	0.6
β_H	1.4	1.4	1.4	1.4	1.4	1.4	1.5	1.5
ψ	0%	4%	7%	2%	4%	9%	10%	19%

Table 2: Leverage-constraints calibration

Table 3

Performance of BAB conditional on volatility.

	Panel A: Whole-sample regressions			Panel B: FF6 by volatility state		
	CAPM	FF3	FF6	Low	High	L-H
Alpha (%)	0.46 (2.49)	0.44 (2.71)	-0.08 (-0.45)	0.56 (2.92)	-0.41 (-1.61)	0.97 (3.03)
MKT	-0.02 (-0.31)	0.16 (3.12)	0.31 (6.62)	0.18 (3.53)	0.44 (6.35)	-0.25 (-2.93)
SMB		-0.66 (-9.01)	-0.59 (-8.52)	-0.59 (-7.57)	-0.60 (-5.77)	0.01 (0.11)
HML		0.33 (3.57)	0.15 (1.53)	0.17 (1.45)	0.06 (0.43)	0.11 (0.60)
RMW			0.32 (3.33)	-0.16 (-1.05)	0.49 (3.94)	-0.65 (-3.34)
CMA			0.62 (4.82)	0.23 (1.37)	0.78 (4.48)	-0.55 (-2.25)
MOM			0.30 (3.67)	0.12 (2.08)	0.37 (3.63)	-0.25 (-2.09)
R ² (%)	-0.14	21.40	34.24	25.33	40.62	
Panel C: Sharpe ratio						
SR				0.72	0.13	0.59
p-value(SR)				-	-	0.040

Table 3: BAB conditional on volatility

5.3 Table 4: conditional factor models

Read:

- Table 4 estimates conditional versions of the CAPM, FF3, FF6, consumption, and intermediary leverage models.
- Lagged volatility continues to negatively predict BAB abnormal returns across models.
- The effect is economically large and survives conditioning on lagged market returns and credit spreads.
- Conditional factor loadings do not eliminate the low-volatility BAB alpha.

Economic message: The volatility puzzle is not just a failure of one static factor model. Even flexible conditional factor models cannot absorb the state dependence.

Table 4
Performance of BAB relative to conditional factor models.

	CAPM		FF3		FF6		Consumption		Leverage	
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)
α_0	0.43 (2.32)	0.41 (2.23)	0.43 (2.60)	0.45 (2.93)	0.38 (1.14)	0.30 (1.91)	0.47 (2.55)	0.51 (2.83)	0.25 (1.35)	0.25 (1.33)
α_0^{vol}	-0.33 (-1.38)	-0.67 (-2.23)	-0.29 (-1.51)	-0.69 (-3.07)	-0.43 (-2.34)	-0.59 (-2.92)	-0.27 (-1.19)	-0.52 (-1.70)	-0.41 (-1.94)	-0.60 (-1.96)
α_0^{DEF}		-0.06 (-0.20)		0.26 (1.16)		0.40 (2.08)		0.20 (0.76)		0.09 (0.33)
α_0^{DP}		-0.31 (-0.98)		-0.31 (-1.33)		-0.16 (-0.63)		-0.44 (-1.36)		-0.27 (-0.81)
α_0^{MAX}		-0.21 (-1.02)		-0.17 (-0.92)		0.02 (0.11)		-0.17 (-0.82)		-0.20 (-0.86)

This table reports α_0 and α_1 estimates from regressions of the form:
 $BAB_t = \alpha_0 + \alpha_1'x_{t-1} + \sum_{i=1}^K (r_{i,t-1} + r_{i,t-2} + \dots + r_{i,t-K})/x_{t-1} + \epsilon_t$
 where f_t is the return on factor $k = 1, \dots, K$. In odd-numbered columns, the vector of instruments, x_{t-1} , only includes the log realized volatility on BAB (Vol). In even-numbered columns, x_{t-1} also includes the difference between BAA and AAA bond yields (DEF), the log dividend-price ratio on the CRSP value-weighted index (DP), and the difference in market betas between the long and short legs of BAB ($Beta$) before holding the market exposure. The instruments are detrended and standardized so that the α_1 coefficients are the impact, in percentage points, of a one standard deviation shock to the instrument. In Columns (1) and (2), "CAPM", the only factor is MKT; in Columns (3) and (4), "FF3", the factors are MKT, SMB, and HML; in Columns (5) and (6), "FF6", the factors are all six in the FF6 model; in Columns (7) and (8), the only factor is the mimicking portfolio for growth in the unfiltered consumption series of *Goetzke (2017)*; and in Columns (9) and (10), the only factor is the mimicking portfolio for intermediary leverage of *Adrian et al. (2014)* (see Section 4.2 for details). Parentheses below point estimates contain t -statistics based on *White (1980)* heteroskedasticity-robust standard errors. The sample period is July 1965 to December 2016.

5.4 Tables 5 and 6: IVOL, mispricing, and lottery demand

Read:

- Table 5 controls for IVOL and mispricing in several ways: dropping high-IVOL stocks, dropping overpriced stocks, dropping overpriced high-IVOL stocks, and sorting on betas orthogonalized to IVOL or mispricing.
- These adjustments often remove the unconditional FF6 alpha.
- However, the low-minus-high volatility alpha remains large, ranging roughly from 0.76% to 1.02% per month.
- Table 6 orthogonalizes beta to lottery-demand proxies SMAX and MAX.

- Conditional alpha remains high: the low-minus-high spreads are 0.86% and 1.00% per month.

Economic message: Limits to arbitrage, mispricing, IVOL, and lottery demand help explain average returns but not the volatility-conditioned beta anomaly.

Table 5
Conditional performance of BAB controlling for idiosyncratic volatility and mispricing.

	Unc.	Low	High	L-H
Panel A: Dropping high-IVOL stocks				
Alpha (%)	-0.00 (-0.02)	0.54 (2.83)	-0.39 (-1.48)	0.93 (2.85)
SR	0.37	0.77	0.13	0.64
p-value(SR)	-	-	-	0.024
Panel B: Dropping overpriced stocks				
Alpha (%)	-0.26 (-1.54)	0.39 (2.00)	-0.62 (-2.33)	1.01 (3.07)
SR	0.20	0.57	0.00	0.57
p-value(SR)	-	-	-	0.045
Panel C: Dropping overpriced high-IVOL stocks				
Alpha (%)	-0.10 (-0.59)	0.54 (2.78)	-0.48 (-1.86)	1.02 (3.15)
SR	0.31	0.74	0.08	0.66
p-value(SR)	-	-	-	0.022
Panel D: Sorting on beta orthogonalized to IVOL				
Alpha (%)	-0.14 (-0.83)	0.45 (2.31)	-0.45 (-1.78)	0.90 (2.82)
SR	0.30	0.61	0.13	0.48
p-value(SR)	-	-	-	0.094
Panel E: Sorting on beta orthogonalized to mispricing				
Alpha (%)	-0.12 (-0.69)	0.39 (1.81)	-0.37 (-1.38)	0.76 (2.20)
SR	0.30	0.71	0.07	0.64
p-value(SR)	-	-	-	0.024

Table 5: IVOL and mispricing controls

Table 6
Conditional performance of BAB controlling for lottery demand.

	Unc.	Low	High	L-H
Panel A: Sorting on beta orthogonalized to SMAX				
Alpha (%)	-0.10 (-0.61)	0.49 (2.39)	-0.37 (-1.49)	0.86 (2.67)
SR	0.33	0.69	0.12	0.58
	-	-	-	0.043
Panel B: Sorting on beta orthogonalized to MAX				
Alpha (%)	-0.21 (-1.28)	0.41 (2.13)	-0.59 (-2.29)	1.00 (3.11)
SR	0.25	0.61	0.04	0.57
	-	-	-	0.046

Table 6: Lottery-demand controls

5.5 Table 7: sentiment and disagreement

Read:

- Table 7 conditions BAB returns on volatility, sentiment, and analyst disagreement.
- BAB alphas remain highest in low-volatility months across sentiment states.
- The same pattern holds across disagreement states.
- This rejects the idea that volatility merely proxies for high sentiment or high disagreement.

Economic message: Sentiment and disagreement may matter for the beta anomaly, but they do not subsume the volatility-conditioning effect.

5.6 Table 8: BAB volatility versus market volatility

Read:

- Table 8 sorts months jointly by market volatility and BAB volatility.
- BAB abnormal returns are strongest when BAB volatility is low, regardless of market volatility.
- The result indicates that the relevant conditioning variable is own-factor volatility, not simply aggregate market volatility.

Economic message: The beta anomaly's conditional performance is tied specifically to volatility in the BAB factor itself.

Table 7

Abnormal returns of BAB in periods of high and low volatility, sentiment, and analyst disagreement.

Panel A: Sentiment				
Sentiment	BAB volatility			
	Low	High	Low–High	Average
Low	0.57 (2.14)	−0.70 (−1.84)	1.28 (2.73)	−0.07 (−0.28)
High	0.56 (1.98)	−0.12 (−0.33)	0.68 (1.51)	0.22 (0.99)
High–Low	−0.01 (−0.03)	0.59 (1.14)	−0.60 (−0.92)	0.29 (0.89)
Average	0.57 (2.91)	−0.41 (−1.59)	0.98 (3.02)	
Panel B: Disagreement				
Disagreement	BAB volatility			
	Low	High	Low–High	Average
Low	0.19 (0.62)	−0.73 (−1.22)	0.92 (1.37)	−0.27 (−0.80)
High	0.45 (1.11)	−0.13 (−0.32)	0.58 (1.01)	0.16 (0.55)
High–Low	0.26 (0.50)	0.60 (0.83)	−0.35 (−0.39)	0.43 (0.97)
Average	0.32 (1.26)	−0.43 (−1.20)	0.75 (1.70)	

We assign each month to one of four regimes ($j = 1, \dots, 4$) according to whether prior-month investor sentiment and realized volatility of BAB (defined in Table 1) are above or below their median values. Panel A reports alphas estimated in each of these regimes from a regression of the form:

$$BAB_t = \sum_{j=1}^4 D_{j,t} (\alpha_j + \beta_j MKT_t + s_j SMB_t + h_j HML_t + m_j MOM_t + r_j RMW_t + c_j CMA_t) + \epsilon_t, \quad (15)$$

where $D_{j,t}$ equals one if month t is in regime j , and zero otherwise. Panel B reports similar estimates as Panel A, but using the time series of beta-weighted analyst disagreement about long-term earnings growth forecasts in lieu of sentiment. Parentheses below point estimates contain t -statistics based on White (1980) heteroskedasticity-robust standard errors. The sample period is July 1965 to December 2016.

remains at least a marginally significant predictor of BAB abnormal returns after controlling for volatility.

Table 8

Abnormal returns of BAB in periods of high- and low- own-factor and market volatility.

Market volatility	BAB volatility			
	Low	High	Low–High	Average
Low	0.56 (2.61)	0.14 (0.38)	0.42 (0.99)	0.35 (1.64)
High	0.40 (0.96)	−0.43 (−1.32)	0.83 (1.57)	−0.02 (−0.06)
High–Low	−0.16 (−0.35)	−0.57 (−1.16)	0.41 (0.60)	−0.37 (−1.08)
Average	0.48 (2.05)	−0.15 (−0.59)	0.63 (1.84)	

This table assigns months to one of four regimes based according to whether prior-month realized market volatility and realized BAB volatility are above or below their median value and presents FF6 alphas of BAB (defined in Table 1) in each of the four regimes. Regression slopes and intercepts are estimated separately in each regime as in Eq. (15). Parentheses below point estimates contain t -statistics based on White (1980) heteroskedasticity-robust standard errors. The sample period is July 1965 to December 2016.

5.7 Tables 9 and 10: institutional demand and counterfactual repricing

Read:

- Table 9 estimates demand for beta by institution type.
- Institutional investors generally prefer high-beta stocks on average.
- Their beta preference falls sharply in high-volatility periods, while households retain a preference for low-beta stocks.
- Table 10 performs a counterfactual repricing experiment dropping institutions whose portfolio betas are most sensitive to volatility.
- In low-volatility states, removing these institutions makes low-beta stocks appreciate by 3.97% and high-beta stocks depreciate by 4.47%, producing an 8.44% low-minus-high price impact.
- The low-minus-high difference between low- and high-volatility states is 6.91%, large enough to explain the conditional BAB alpha spread.

Economic message: Institutional demand dynamics provide a mechanism of the correct sign and magnitude: institutions retreat from high-beta stocks when volatility rises, affecting relative valuations of high- and low-beta portfolios.

Table 9
Demand system coefficients by institution type.

	Households	Banks	Insurance companies	Investment advisors	Mutual funds	Pension funds	Other	All institutions
Unc.	-0.10 (-6.42)	0.06 (4.19)	0.08 (3.51)	0.06 (3.01)	0.10 (7.23)	0.16 (7.20)	0.19 (4.75)	0.09 (5.93)
Low vol	-0.10 (-4.30)	0.12 (4.76)	0.14 (3.33)	0.19 (5.61)	0.19 (6.99)	0.30 (7.21)	0.33 (4.45)	0.18 (6.43)
High vol	-0.11 (-4.76)	-0.00 (-0.03)	0.03 (1.20)	-0.06 (-2.57)	0.03 (2.11)	0.04 (1.77)	0.08 (1.81)	0.01 (0.55)
Low-High	0.00 (0.10)	0.13 (4.28)	0.11 (2.42)	0.25 (6.11)	0.16 (5.21)	0.26 (5.46)	0.25 (2.92)	0.17 (5.41)

For each institution-quarter, (i, t) , we estimate the following demand system for portfolio weights via GMM:

$$E(v_{i,t}(n)|\hat{\theta}_{i,t}(n), \beta_{i,t}(n), x_{i,t}(n)) = 1, \quad (21)$$

$$u_{i,t}(n) = \exp(\theta_{i,t}^T \text{vec}(n) + \theta_{i,t}^T \beta_{i,t}(n) + \theta_{i,t}^T x_{i,t}(n) + \theta_{i,t}^T \epsilon_{i,t}(n)), \quad (22)$$
where $n = 1, \dots, N_t$ indexes stocks that exist at time t , and $u_{i,t}(n)/u_{i,t}(0)$ is the portfolio weight in stock n normalized by the weight in the outside asset. This table reports time-series means of quarterly assets under management-weighted averages, $\hat{\theta}_{i,t}^T$, of the parameter $\theta_{i,t}^T$ over the full sample (row “Unc.”), the subsample with below-median realized volatility (“Low vol”), and the subsample with above-median volatility (“High vol”), as well as low-minus-high difference (“Low-High”). Parentheses below point estimates contain t -statistics based on White (1980) heteroskedasticity-robust standard errors. The sample period is 1980q1 to 2016q4.

Table 9: Institutional demand coefficients

Table 10
Counterfactual repricing statistics.

	$PI(L)$	$PI(H)$	Low-High
Unc	2.21 (4.41)	-2.02 (-3.27)	4.23 (3.98)
Low vol	3.97 (5.45)	-4.47 (-6.30)	8.44 (6.20)
High vol	1.07 (1.66)	-0.45 (-0.52)	1.53 (1.06)
Low-High	2.90 (2.98)	-4.01 (-3.59)	6.91 (3.49)

Based on the estimated demand system defined in Table 9, each quarter, we compute the price impact statistics, defined in Eq. (24), of the low- and high-beta portfolios (“ $PI(L)$ ” and “ $PI(H)$,” respectively) in BAB from a counterfactual experiment. In this experiment, we drop institutional investors with the greatest sensitivity of portfolio betas with respect to volatility, estimated via Eq. (32), and then reallocate their assets to all remaining institutions. Each panel reports time-series means of these price impact statistics (in units of %) over the whole sample (“Unc”), months with below-median BAB volatility (“Low vol”), and months with above-median volatility (“High vol”). Parentheses below point estimates contain t -statistics based on White (1980) heteroskedasticity-robust standard errors. The sample period is 1985q1 to 2016q4.

Table 10: Counterfactual repricing

6 Short summary

- The paper studies the beta anomaly through the conditional performance of BAB.
- Volatility-managed BAB has a higher Sharpe ratio than unmanaged BAB.
- BAB alpha and Sharpe ratio are much higher after low-volatility months.
- Leverage-constraints theory predicts the wrong volatility-Sharpe relation.
- FF6 prices BAB unconditionally but not conditionally.
- Conditional factor models do not solve the puzzle.
- IVOL, mispricing, lottery demand, sentiment, and disagreement do not explain the low-volatility alpha.
- Institutional investors prefer high-beta stocks on average but reduce that preference when volatility rises.

- A counterfactual demand experiment generates low-minus-high beta price impacts large enough to explain the conditional BAB alpha.
- The paper concludes that time-varying institutional demand is central to the volatility puzzle of the beta anomaly.

7 Conclusion

7.1 Core conclusion

Barroso, Detzel, and Maio show that the beta anomaly remains unexplained when evaluated through conditional performance. Leading explanations can match unconditional BAB returns but fail to explain why BAB is most profitable and most anomalous after low-volatility months.

7.2 Economic intuition

The core intuition is that low-beta stocks become especially underpriced relative to high-beta stocks in low-volatility states. Existing theories such as leverage constraints, mispricing, lottery demand, sentiment, and disagreement do not generate the right state dependence. Institutional investors' demand does: when volatility rises, institutions reduce demand for high-beta stocks, and the price impact of this behavior changes relative valuations across volatility states.

7.3 Takeaway

The paper's broader lesson is methodological and economic. Methodologically, anomaly theories must be tested conditionally, not only unconditionally. Economically, the beta anomaly appears tied to time-varying institutional demand and portfolio-management incentives, not simply to a static risk, arbitrage, or behavioral mechanism.